Electromagnetic Induction

Exam code: 0625 & 0972

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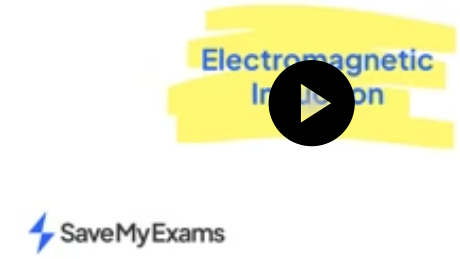

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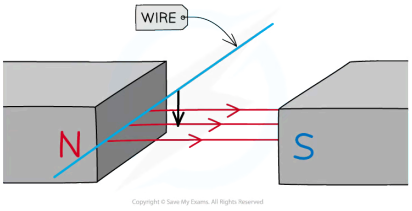
Help me understand

Induced e.m.f.

- An electromotive force (e.m.f.) is induced in a conductor whenever there is **relative movement** between the conductor and a magnetic field
- This could be when
  - the conductor moves in a stationary magnetic field
  - the conductor is stationary in a changing magnetic field

Induced e.m.f. due to a moving conductor

- For an electrical conductor moving in a fixed magnetic field:
  - the conductor (e.g. a wire) **cuts** the field lines
  - an e.m.f. is **induced** in the wire



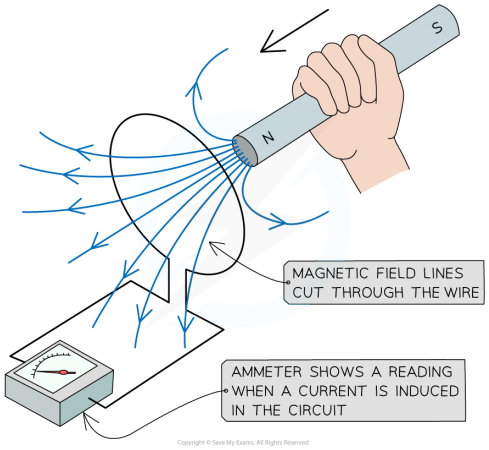
When an electrical conductor moves in a magnetic field an e.m.f. is induced

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- For a fixed conductor in a changing magnetic field:
  - as the magnet moves through the conductor (e.g. a coil), the field lines **cut** through the turns on the conductor (each individual wire)
  - an e.m.f. is **induced** in the coil



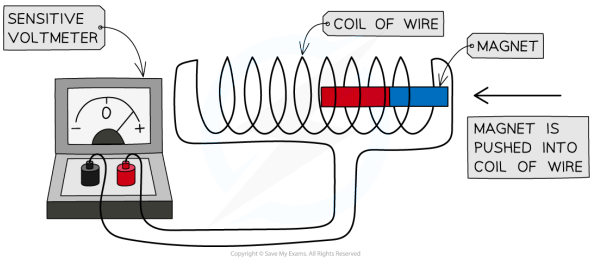
When a magnet is moved towards a wire, the changing magnetic field induces a current in the coil of wire

- A **sensitive voltmeter** can be used to measure the size of the induced e.m.f.
- If the conductor is part of a **complete circuit** then a **current** is induced in the conductor
  - This can be detected by an ammeter

Worked Example

A coil of wire is connected to a sensitive voltmeter.

When a magnet is pushed into the coil the needle on the voltmeter will deflect to the right as shown in the diagram below.



What will happen to the pointer on the voltmeter when the magnet is stationary in the centre of the coil?

A The needle will deflect to the left

D The needle will deflect to the left and then to the right

ANSWER: C

- There is no relative movement between the coil and the magnetic field when both the magnet and coil are stationary
- Since no magnetic field lines are being cut, no e.m.f. will be induced
  - Therefore, the needle will **not** deflect

- **A**, **B** & **D** are incorrect because a deflection on the voltmeter would indicate that an e.m.f. has been induced
- This would only happen if there was relative movement between the coil and the magnetic field

Did this video help you? Yes No

Lenz's law

Extended tier only

- Lenz law states:

The direction of an induced emf always opposes the change causing it

- This means that any magnetic field created by an induced emf will act so that it tries to stop the wire or magnet from moving

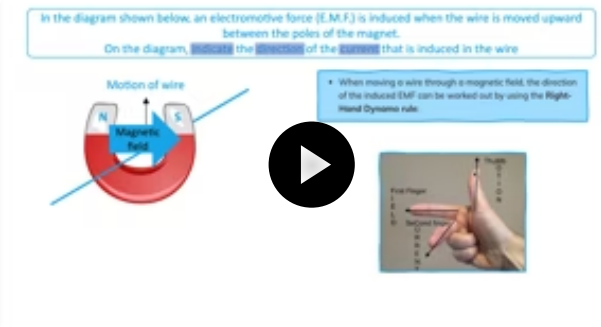
Demonstrating Lenz's law

- Lenz's law can be demonstrated when a magnet is pushed into, or out of, a coil of wire
- If the magnet is pushed north end first into the coil, the end of the coil closest to the magnet will become a **north pole**
- This happens because:
  - the changing magnetic field induces an **emf** in the coil
  - the induced emf causes a current to flow and generates a magnetic field in the coil
  - the magnetic field due to the current **opposes** the magnet being pushed into the coil

When a magnet is pushed into a coil of wire, the end of the coil closest to the magnet will become a north pole and oppose its motion

- If a magnet is now pulled away from the coil of wire, the end of the coil closest to the magnet will become a **south pole**
- This happens because:
  - the changing magnetic field induces an **emf** in the coil
  - the induced emf causes a current to flow and generates a magnetic field in the coil
  - the magnetic field due to the current **opposes** the magnet being pulled away from the coil
  - therefore, the end of the coil closest to the magnet becomes a **south pole**
  - this means it **attracts** the north pole of the magnet

When a magnet is pulled away from a coil of wire, the end of the coil closest to the magnet will become a south pole and oppose its motion



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Extended tier only

- When moving a wire through a magnetic field, the direction of the induced emf can be determined using the **right-hand dynamo** rule
- **F**irst **F**inger = **F**ield:
  - Start by pointing the **f**irst **f**inger (on the right hand) in the direction of the **f**ield
- Thu**M**b = **M**otion:
  - Next, point the thu**m**b in the direction that the wire is **m**oving in
- Se**C**ond = **C**urrent:
  - The Se**c**ond finger will now be pointing in the direction of the **c**urrent (or, strictly speaking, the emf)

The right-hand dynamo rule can be used to deduce the direction of the induced emf

Examiner Tips and Tricks

Remember that current is always in the direction of positive charge carriers. Therefore, current flows from the positive to the negative terminal of the battery.

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